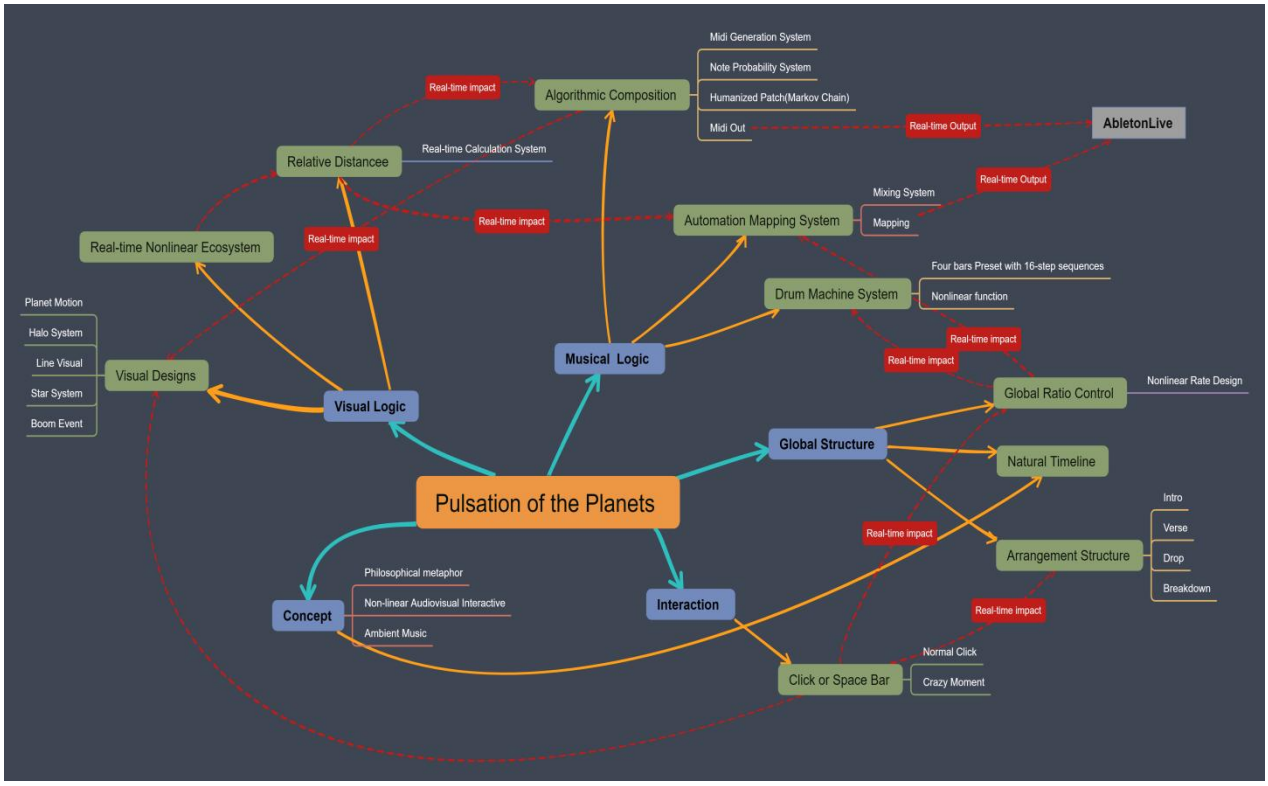
Pulse of the Planets

Brief Introduction

The Pulsation of the Planets is an Audiovisual Interactive work that Generates Real-time Ambient Music through Algorithmic Composition.Create by MaxMSP & Ableton live.
The piece metaphorically reflects the pulsation of planets—how distance, motion, and interaction give Music to this Universe.
However, due to the limited space and time, it might be difficult to provide a detailed technical introduction.

Project Schematic Diagram



Operating instruction

In this work, you can interact with the universe by Manually Triggered(press the Space Bar or Click the Button), and this determines the progress of the real time music you hear.In the Demo Video, I perform the piece through a pre-designed musical structure:

Intro – Verse – Highlight Moment(Manually Triggered) – Drop – Verse – Highlight Moment - Drop – Verse - Ending.

Of course, you can also let it naturally transition from the intro into the verse section, where the verse will evolve and transform on its own.

Background note

Newton's law of universal gravitation and Kepler's laws of planetary motion state that the speed of a planet's movement mainly depends on:

▪ Mass of the central celestial body: For instance, the greater the mass of the sun, the stronger the gravitational force, and the faster the planets need to move in order to maintain a stable orbit.

▪ The distance between a planet and the central celestial body: The closer it is to the center, the stronger the gravitational force and the faster the speed; the farther it is, the slower the speed.

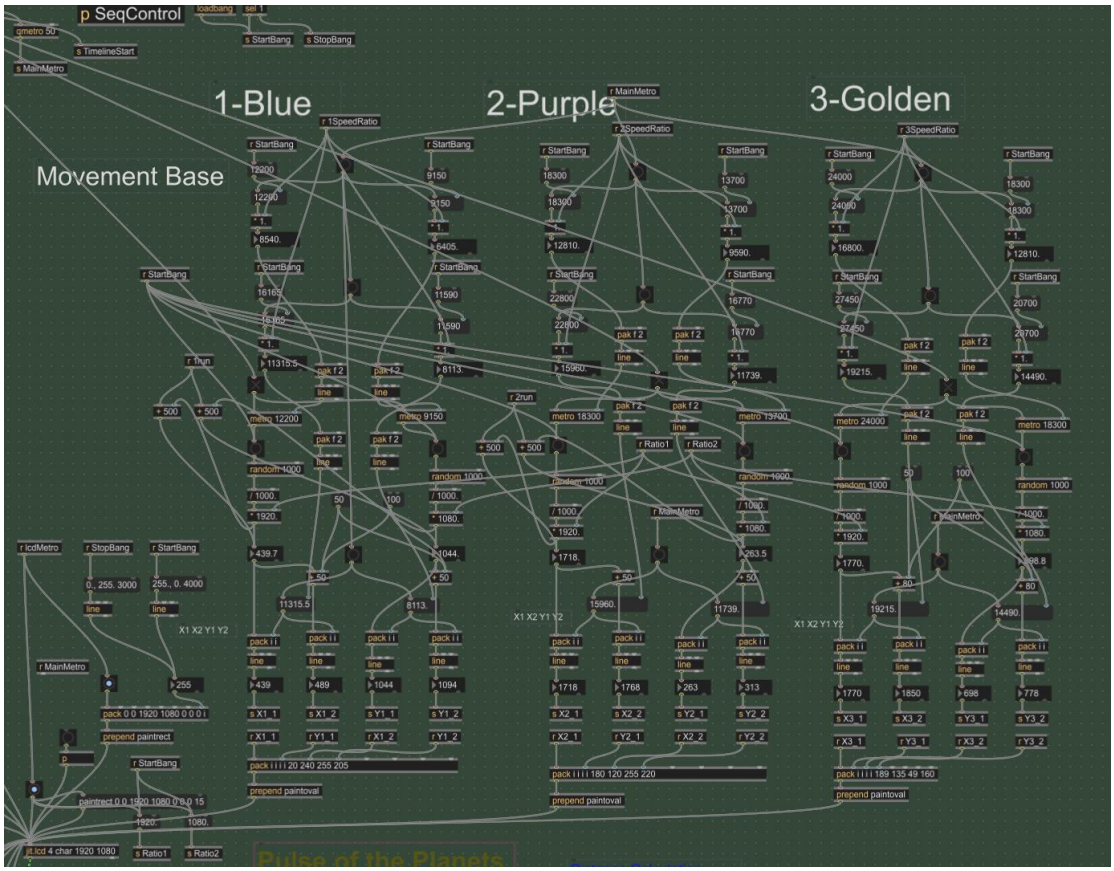
This gave me an inspiration when I was initially creating the work: When the Non-linear Tempo and harmony degree of Ambient Music is combined with the Distance and Relationship between the harmonious planets, it may be very interesting.And What I aim to create is not merely an algorithmic composition script, but rather a more like Intuitive Real-time Music Generation and Mixing System , based on the visual cues provided by the visuals.

Technical Realization

Basic Visual

So based on my conception, the moving objects in the images within this system represent the first step in achieving it, as they determine the logic for generating the notes and the real-time changes - in Ambient Music, the changes brought about by automation are an important part of the framework for ambient music composition.

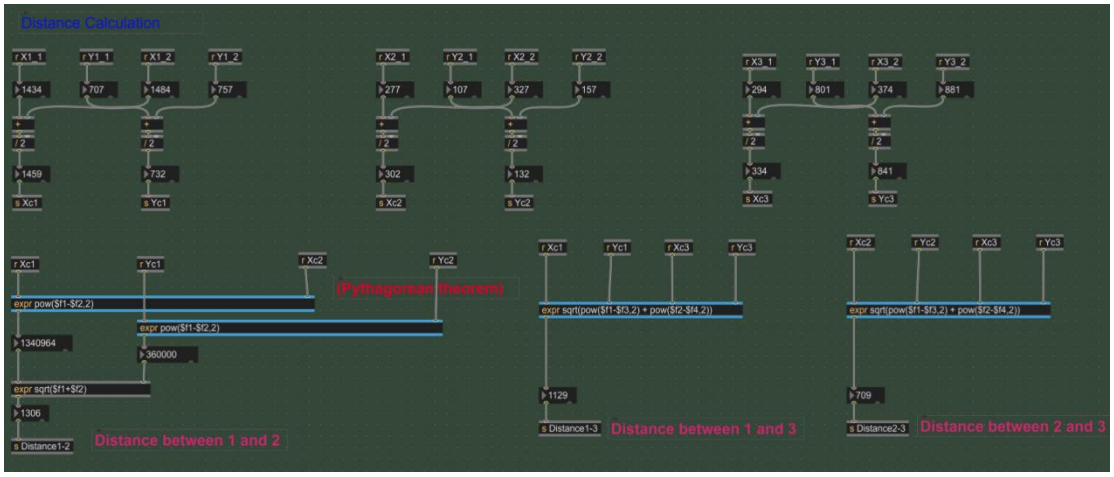
This is roughly the Basic Movement Logic of the three objects



This enables the three objects to randomly update a target point within the screen at intervals that are randomly determined, and they can change to a new random target point at any point during their movement, creating a relatively smooth and non-linear sense of free movement.

This also means that I have obtained three sets of real-time changing xy value data, which forms the basis for the entire music generation and the mapping of musicality.At the same time, I set a **Global Control Ratio**, which enabled the overall real-time nonlinear adjustment mentioned later to be achieved.

Real-time distance detection calculation



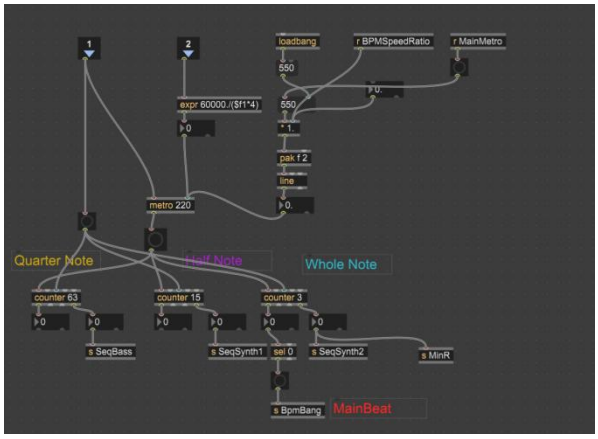
The center coordinates of each object were obtained through basic calculation, allowing precise positioning and packaged them for later calls.Then, the distance between each pair of points was calculated(first by obtaining the differences in their x and y coordinates, and then deriving the straight-line distance using the Pythagorean theorem).I believe there is no need for a detailed explanation.

(I made two different implementation methods. The formulas are the same, but the second one is more concise and effective.)

Musical logic

In the music system, although non-linear ambient music is my goal, but for the sense of organization, I built a Main Metre to organize the musical structure.

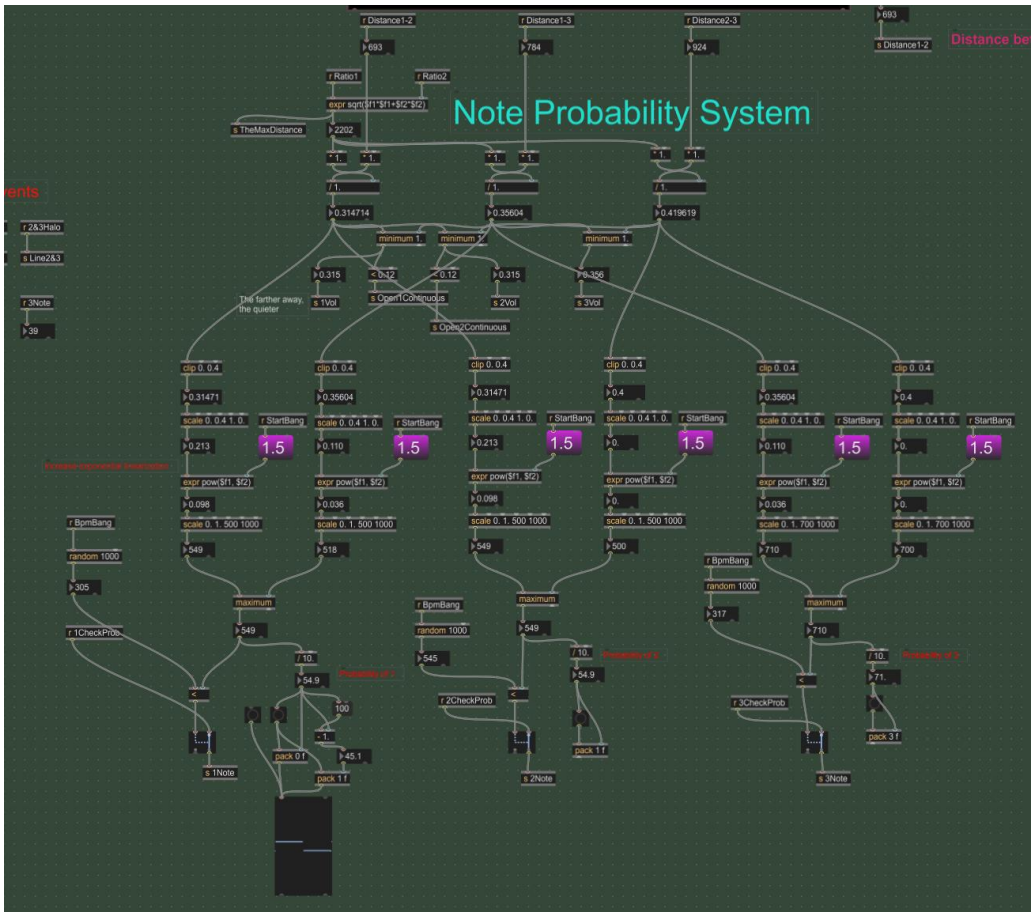
Musical Main Metre



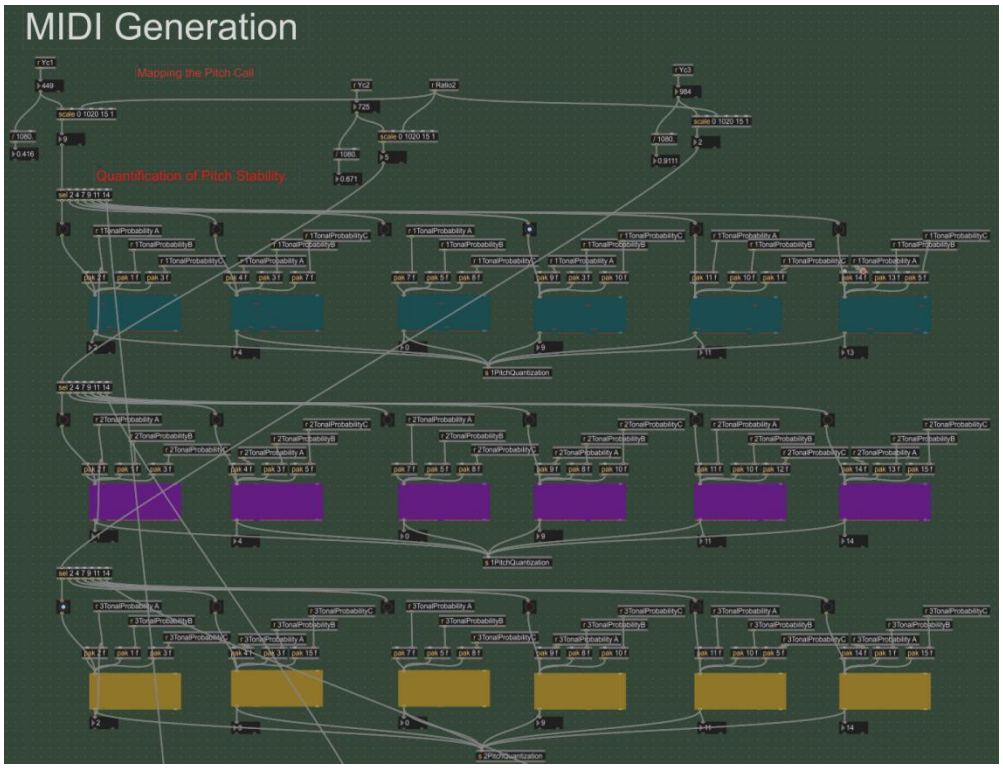
The quantification time value of each object is different.The blue one (The following is represented by object A) only responses in the beat of Quarter Note.The purple one(B) only responses in the beat of Half Note, and the golden one(C) only in Whole Note.So there ratio is **1:4:16**，Output at the Counter 4, 16 and 64 respectively.

Music Create System

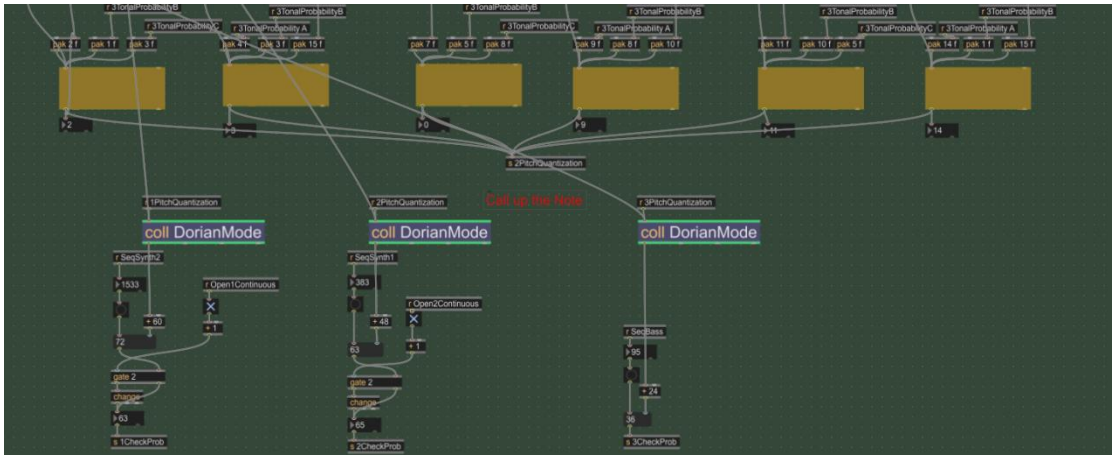
Note Probability System



Midi Generation System (Include Mapping the Pitch Call)



Call up Notes



The Midi generation system is composed of multiple modules：

- **Note Probability System**
- **Midi Generation System**
- **Mapping the Pitch Call&Call up Notes**

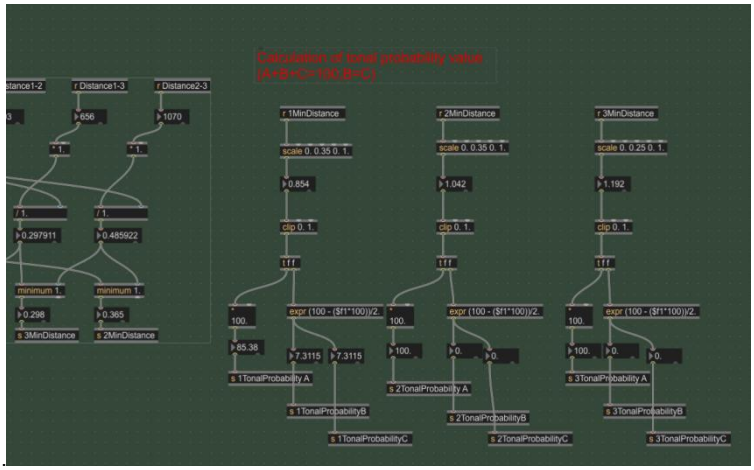
The first part **Note Probability System** is essentially a filtering of the rhythm (the density of note triggers)Building on the previously calculated distance data among A, B, and C, For each frame,We first Clip distances by a resolution range; then apply a power-curve dynamic shaping; the shaped value is mapped to a probability and compared with random(1000) to trigger by probability.

At the same time, a module capable of triggering the same sound continuously will be activated at a certain distance.

In the second stage **Midi Generation System,** performs probabilistic quantization on the notes themselves at first. Notes are first selected through the base coll index, then pass into a second probabilistic layer of modal stability.As distance decreases, unstable tones are more likely to be corrected toward other stable tones in this modes.

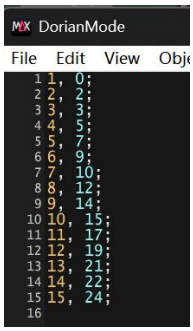
These probabilities are continuously mapped to **[Itables]**, enabling subtle real-time pitch quantization.

Probability ratio calculation for the table.



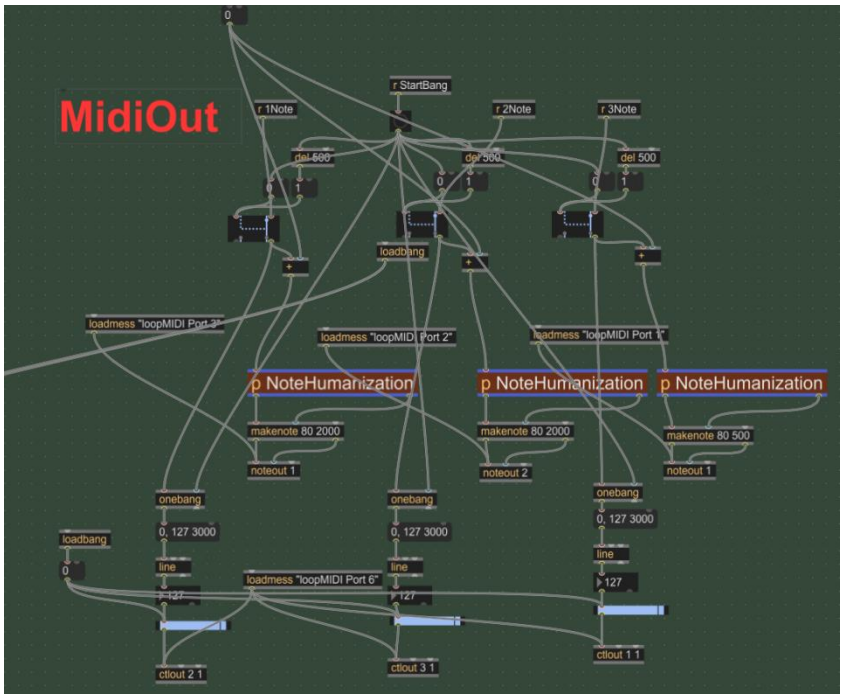
At the same time, some minor adjustments were made to the bass layer (C). In terms of arrangement, the Bassline requires greater stability

Finally, collect the values from the coll file through the index.Using the "coll" , I set the base mode index to the Dorian mode.

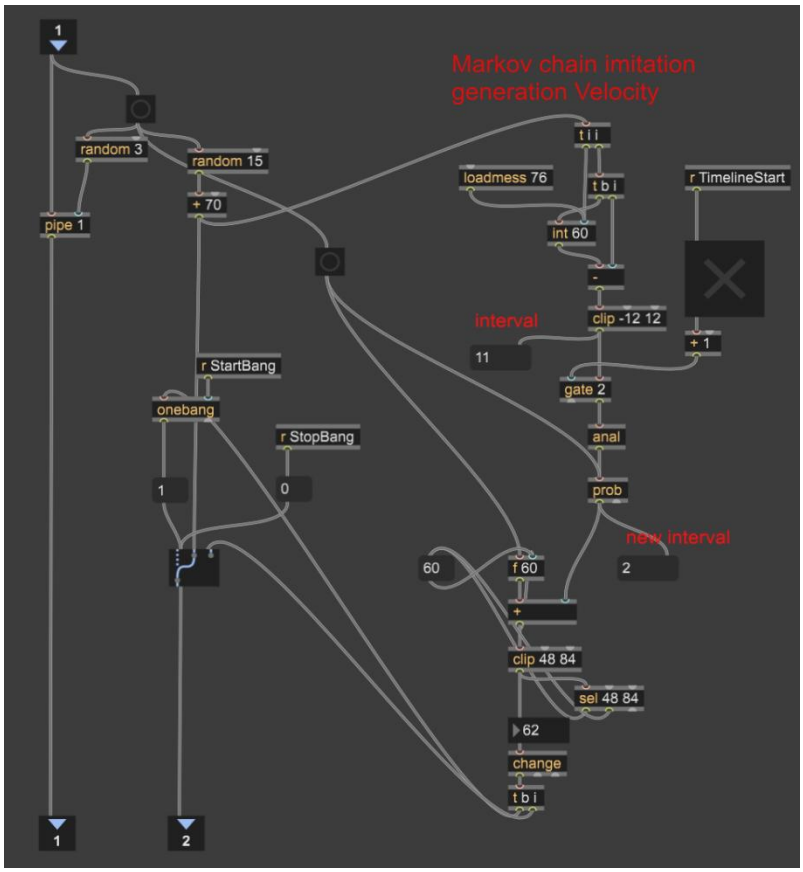


And then use an external root offset to shift the pitch region.This allows the same scale structure to be layered across different octaves—creating separate tonal ranges for two synths(A,B) and bass(C) parts.

Midi Out to AbletonLive



Humanized Patch



Before sending MIDI output, I designed a small patch that applies subtle humanization combined with a Markov chain to learn and generate velocity variations.

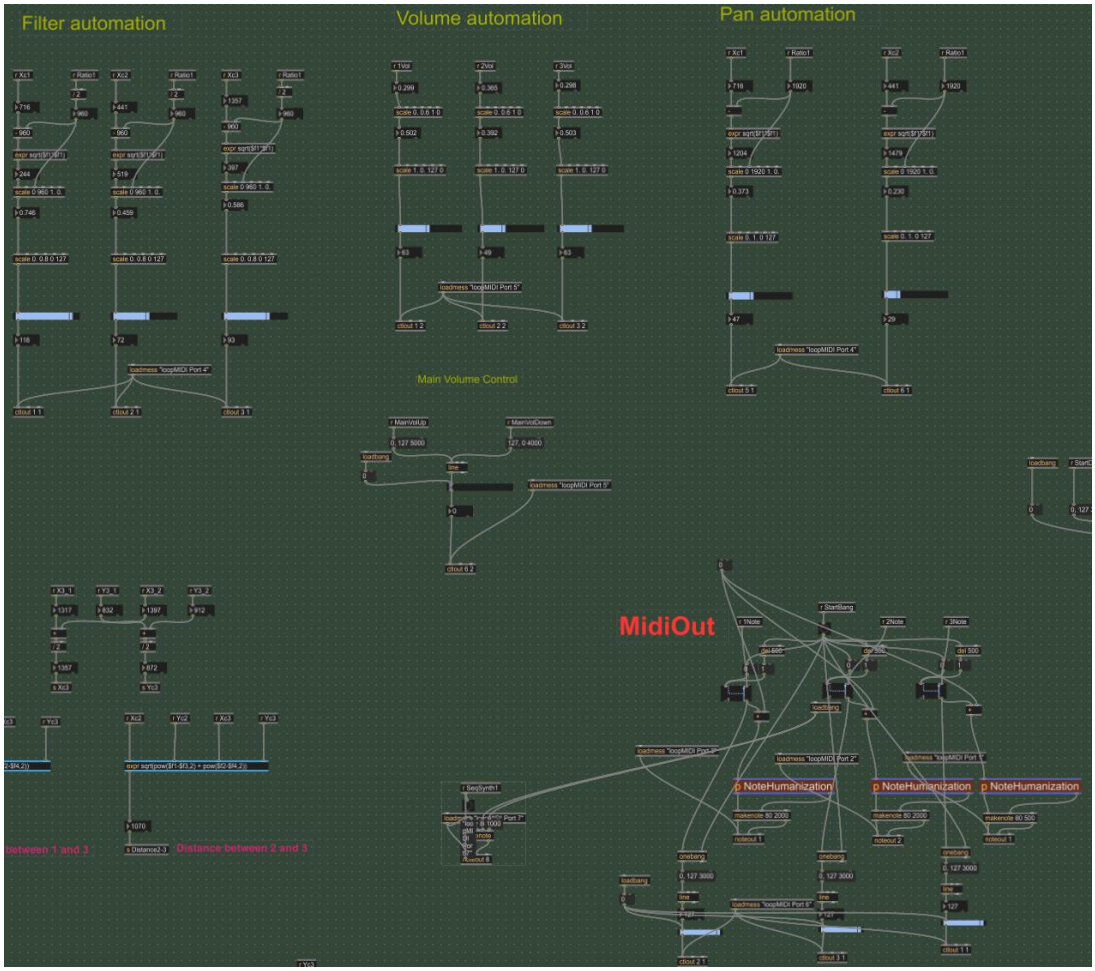
As pitch learning did not sound fitting for this ambient piece, the Markov process is applied only to velocity.

Mapping within Ableton



通.	音符/控制	路径	名称	最小	最大
1	CC 1	2-Glass Detuned Fast Attack Pad | Pro-Q 4	Band 1 Gain	-30.00 dB	+1.00 dB
1	CC 1	BASS | Mixer	Track Volume	-24.0 dB	-13.0 dB
2	CC 1	2-Glass Detuned Fast Attack Pad | Limiter	Input Gain	-10 dB	3.0 dB
1	CC 2	2-Glass Detuned Fast Attack Pad | Mixer	Track Volume	-18.0 dB	-2.2 dB
1	CC 2	3-Glass Detuned Fast Attack Pad | Pro-Q 4	Band 1 Gain	-30.00 dB	+1.00 dB
2	CC 2	3-Glass Detuned Fast Attack Pad | Limiter	Input Gain	-10 dB	3.0 dB
1	CC 3	3-Glass Detuned Fast Attack Pad | Mixer	Track Volume	-19.0 dB	-2.9 dB
1	CC 3	BASS | Pro-Q 4	Band 1 Gain	-30.00 dB	+1.00 dB
2	CC 3	BASS | Limiter	Input Gain	-8.0 dB	2.0 dB
1	CC 5	2-Glass Detuned Fast Attack Pad | dearVR MICRO	Azimuth	-150.0 °	150.0 °
1	CC 6	3-Glass Detuned Fast Attack Pad | dearVR MICRO	Azimuth	-66.0 °	66.0 °
2	CC 6	Main | Mixer	Track Volume	-∞ dB	-0.1 dB
1	CC 7	5-Halfstep Kit | Mixer	Track Volume	-∞ dB	-2.0 dB

Automation Mapping System

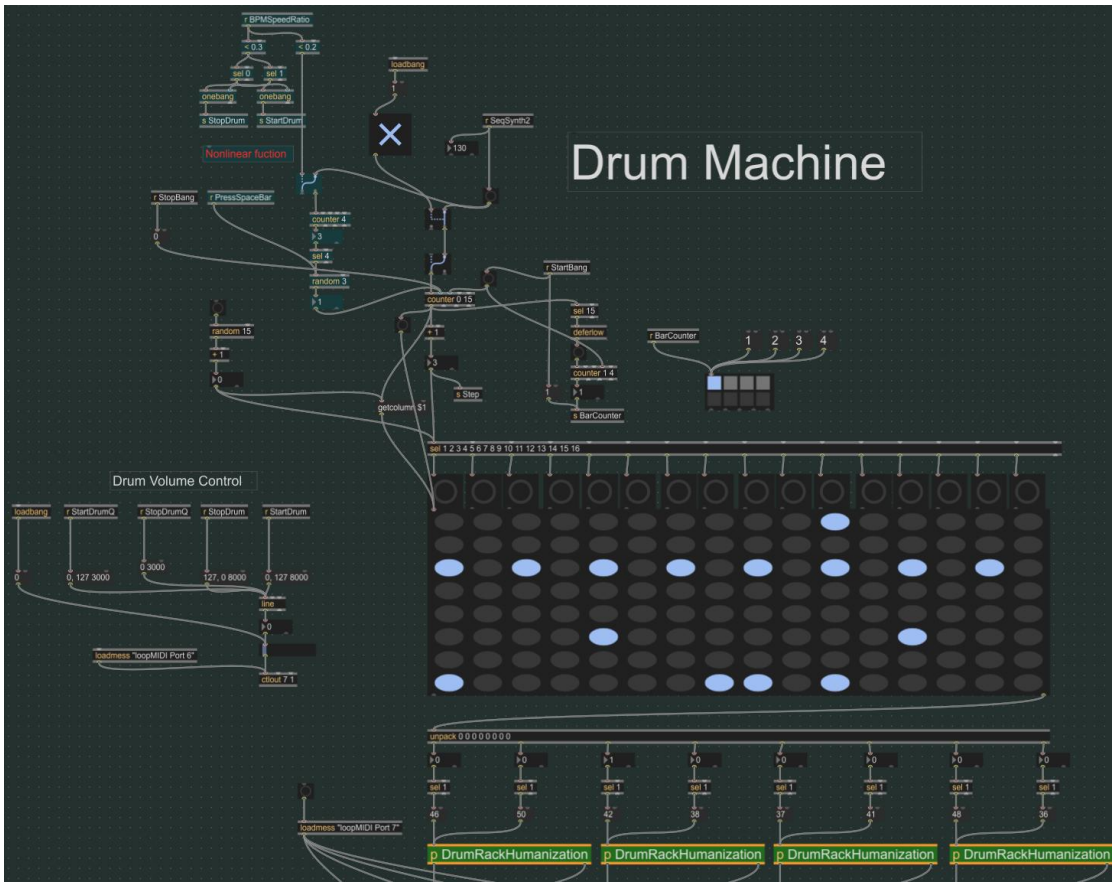


Through mapping, A,B and C achieved the corresponding automated effect:

As each object moves closer to the center of the screen, a shelving filter gradually releases its high-frequency attenuation, resulting in a brighter timbre.Conversely, when the object moves away, the sound becomes darker.At the same time, the overall volume increases as the absolute distance between objects decreases, and diminishes as they move apart.

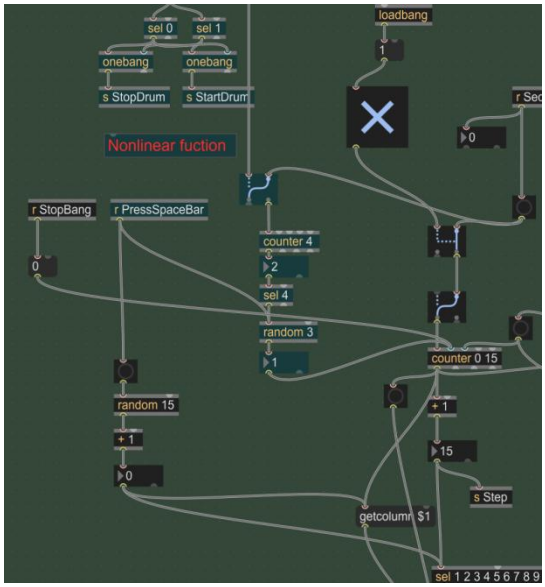
Each object's horizontal position on the screen is also mapped to the stereo panorama, determining its spatial placement.

Drum Machine System



Using **[matrixctrl]** and **[preset]** objects, I simulated a drum machine that stores four bars with 16-step sequences.The system will be activated when the **Global Control Ratio** reaches a certain level (this parameter is a crucial global control parameter and will be discussed later)

Nonlinear function

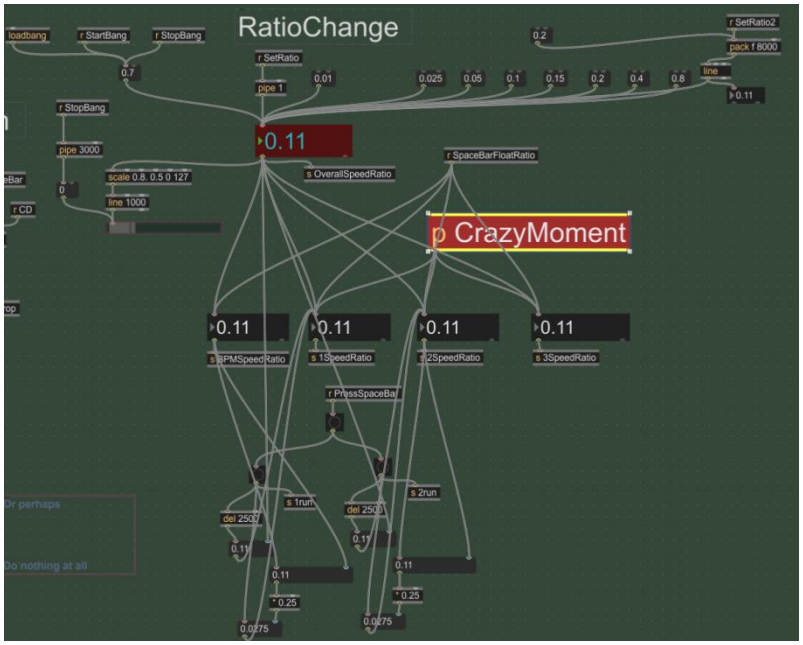


It is worth noting that here, the first cold interface of the counter is utilized: by randomly switching among three playback modes at any time, non-linear drum loop will create. This function will be activated when the overall speed reaches a certain level, bringing about a dramatic dynamic change.

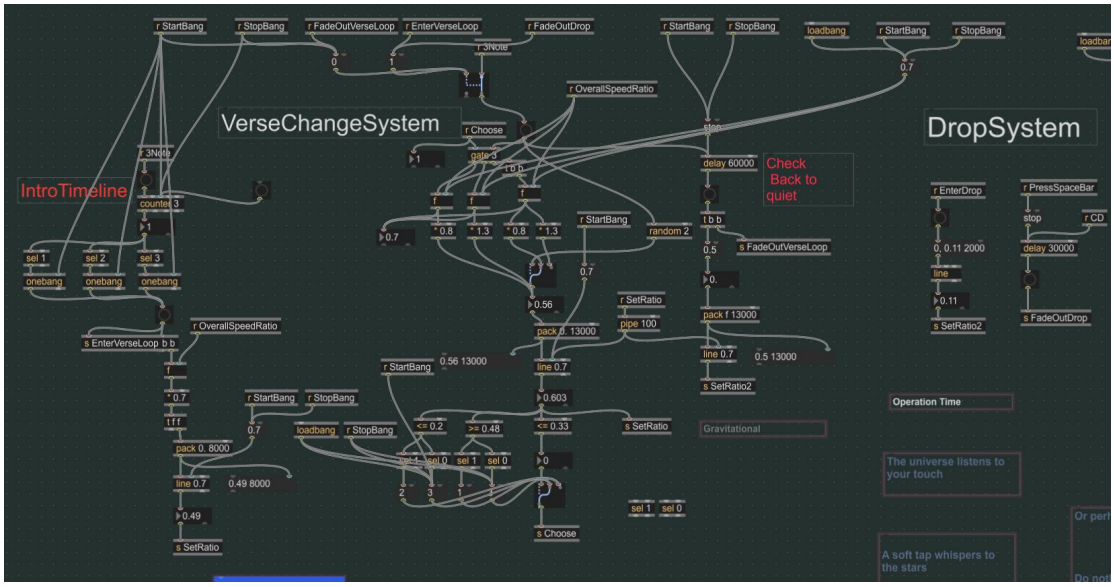
Natural Timeline and Arrangement Structure

At the beginning of the startup process, the overall control rate will be at a lower speed ratio.

Ratio control



Arrangement Structure



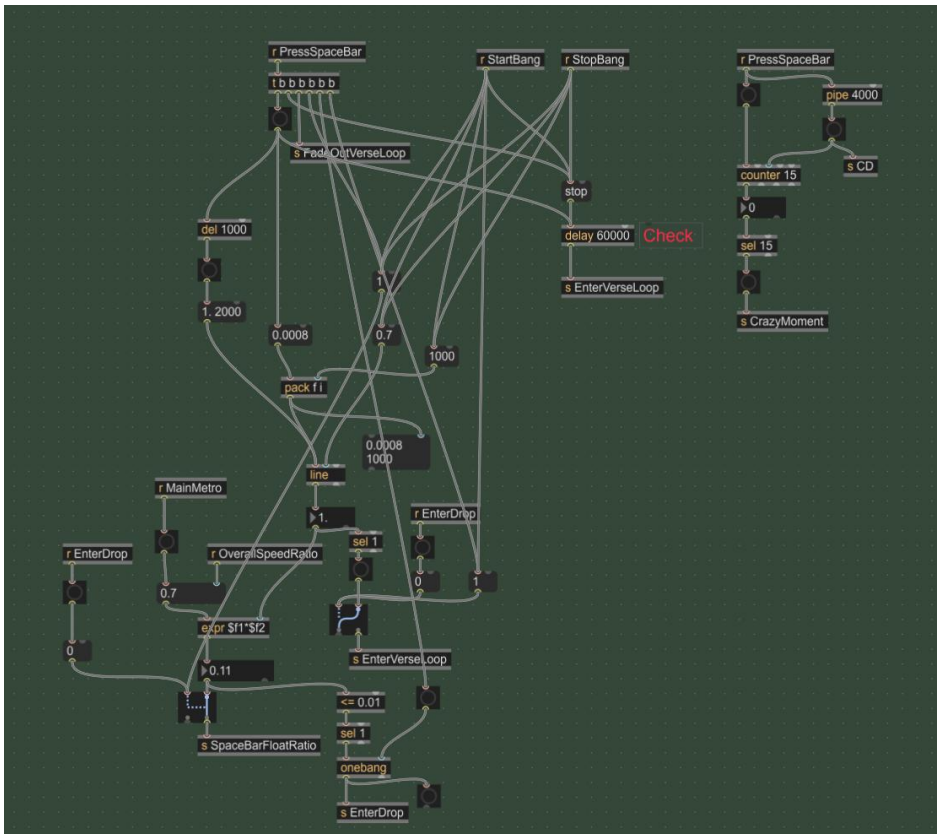
After the system starts, the **C** object determines the transition by probabilistically triggering notes.During its first three successful triggers, the ratio value is non-linearly accelerated through multiplicative scaling.After the third trigger, the system enters the verse section, where the rate continues to change by multiplicative at random intervals, but remains confined within a defined speed range.

If not controlled(Click or Space Bar), this would be the trend of the whole timeline.

But if you can’t resist pressing the Space bar or Click, the Drop section will be triggered after a mysterious event.

It brings a surge of energy—introducing a stable BPM at the peak of excitement, activating a nonlinear drum pattern, and unleashing a burst of flickering starlight across the screen.

Click or Space Bar (Interactive Design)

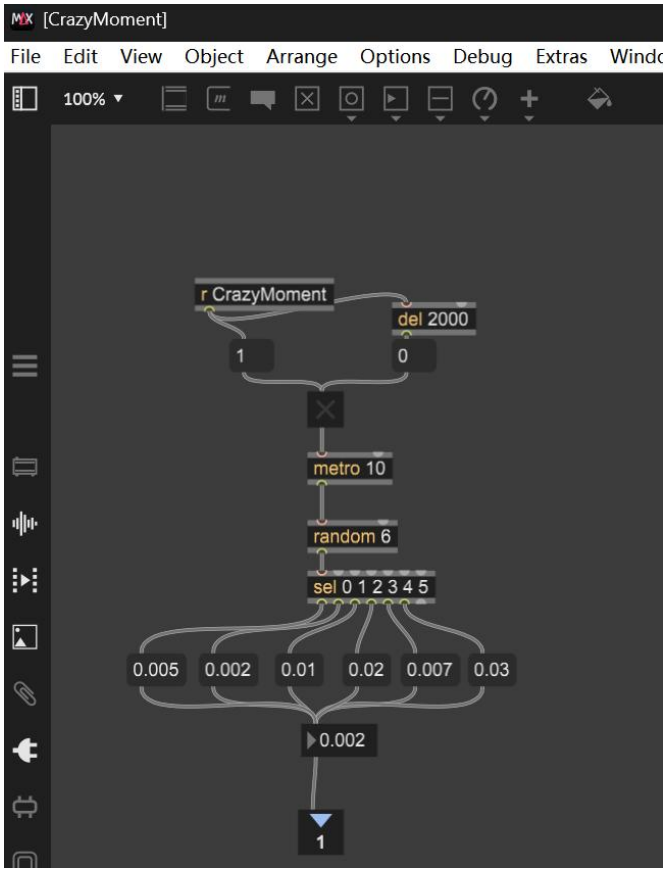


I designed a button that influences the evolution of the universe.It introduces an interactive mechanism: when clicked, the global rate ratio linearly decreases (thus accelerating the overall motion) until it reaches a lower threshold.

During a single click, the value begins to rebound before reaching the minimum, but within 0.8 seconds of consecutive clicks, the rebound is delayed—due to the del mechanism—until no further clicks occur in that window.

By exploiting the behavior of **delay** and **pipe**, I also created a feature called **“Crazy Moment.”**When the button is pressed **15 times within 4 seconds**, a chain reaction is triggered—causing all global parameters to multiply exponentially at high speed, resulting in a chaotic and visually explosive effect.

The Space bar and Click actions serve as key behaviors that can break through the ratio threshold and force the system into the drop section, regardless of whether the Crazy Moment is triggered.



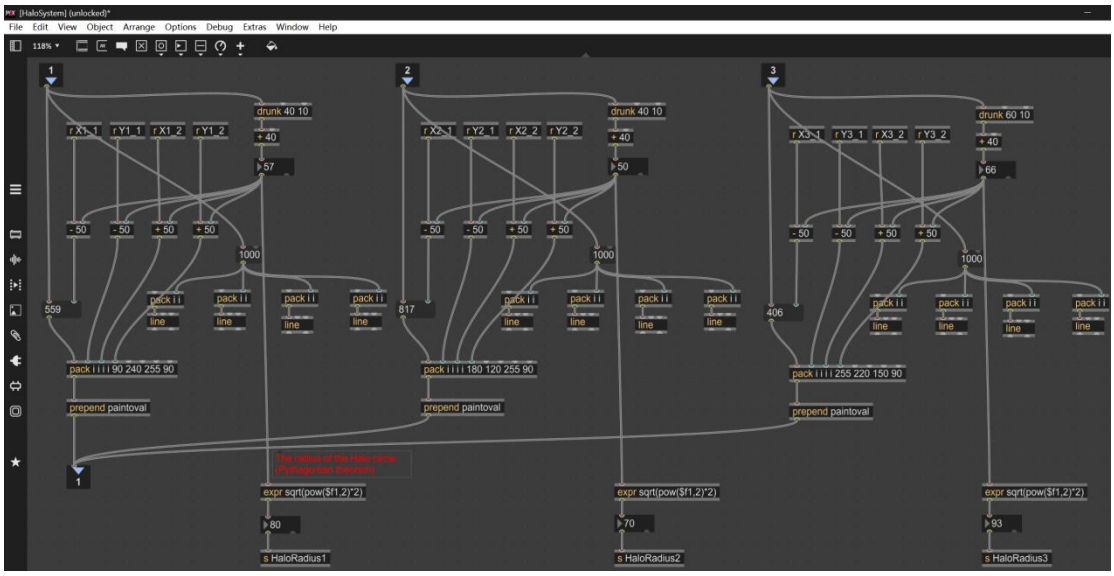
After any interactive action, when the system returns to the verse section, a period of inactivity will gradually slow the global rate down.

This state mirrors the breakdown phase in musical composition—where intensity dissipates—and metaphorically reflects the rhythm of the universe itself:

Everything rises, reaches its peak, and eventually fades away

Some Visual Designs

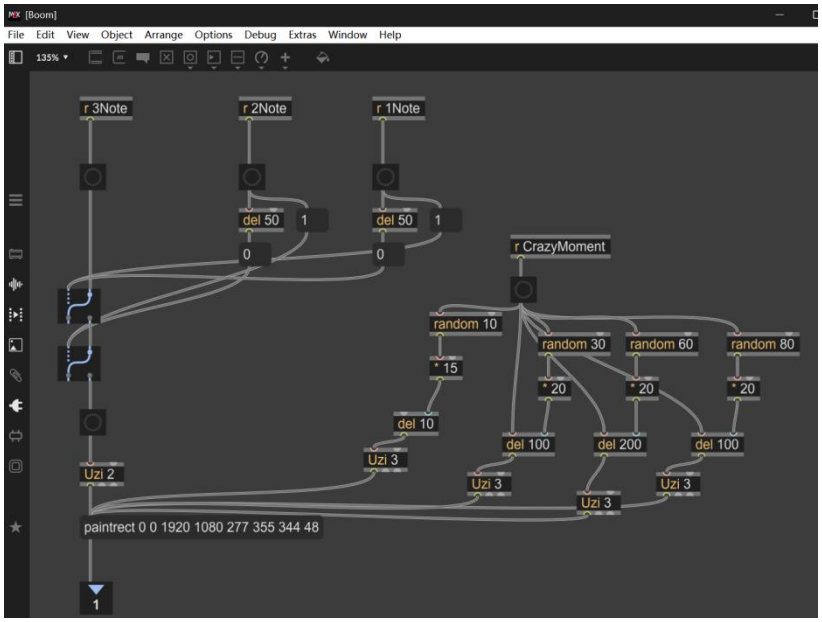
Halo System



At every moment a note event is triggered, a halo system activates—generating a semi-transparent ring with a randomized size.This visual feedback enhances the sense of interaction between sound and image, making each note trigger more tangible and perceptible.

The halo system is also used to detect collision distances.When an interaction occurs, a ray is dynamically emitted between the two objects, visually representing their connection at the moment of contact.

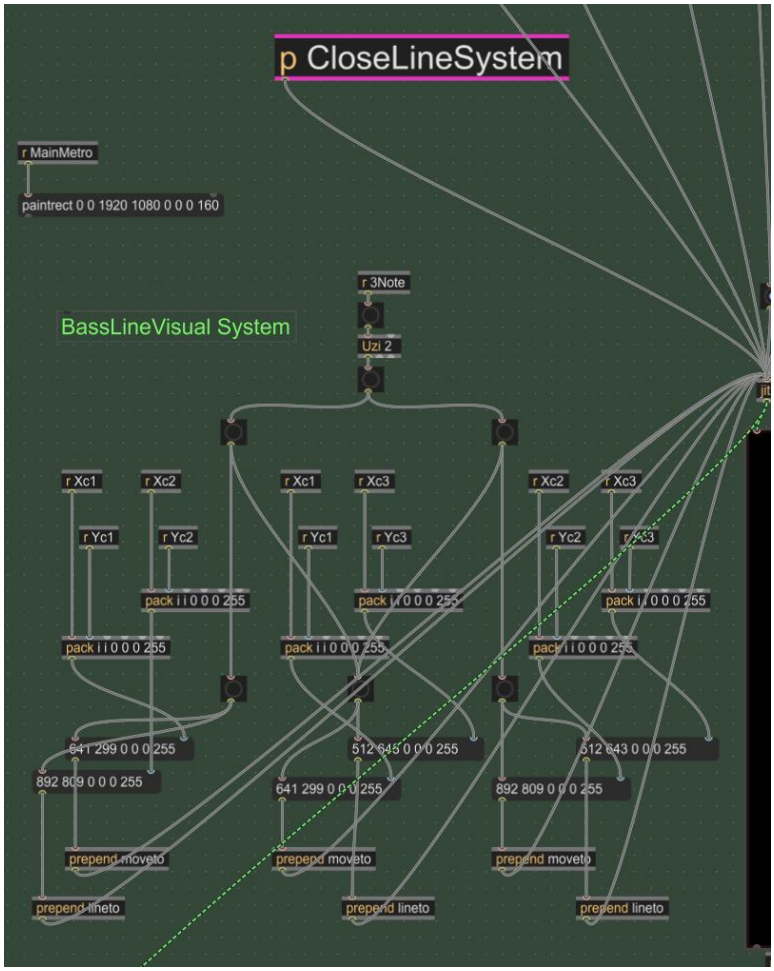
Boom Event



If all three halos trigger within the same beat, a full-screen flash is activated—an event of high randomness that adds visual excitement and structural variation.

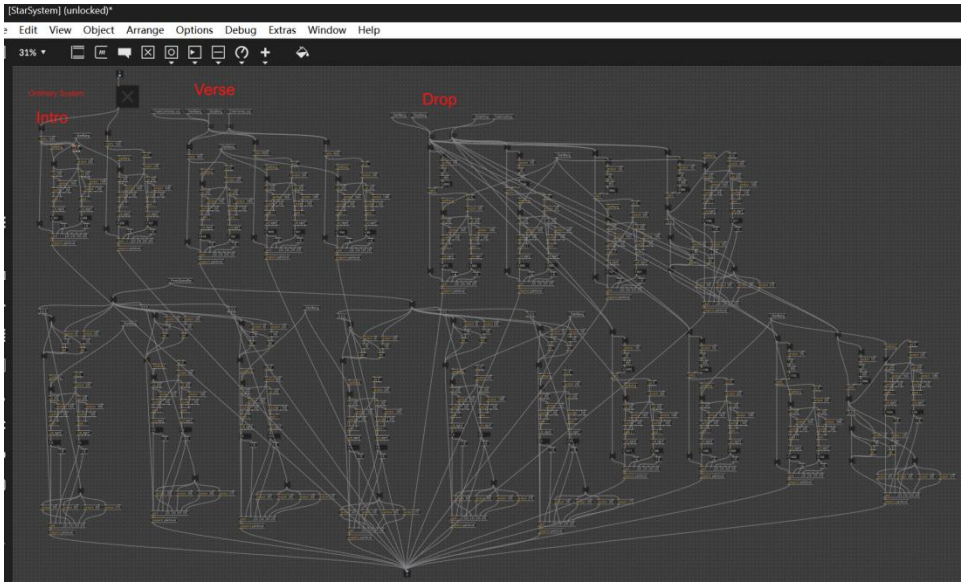
Faster global speeds increase the likelihood of such coincidences, a principle also applied across other system modules.

Line Visual

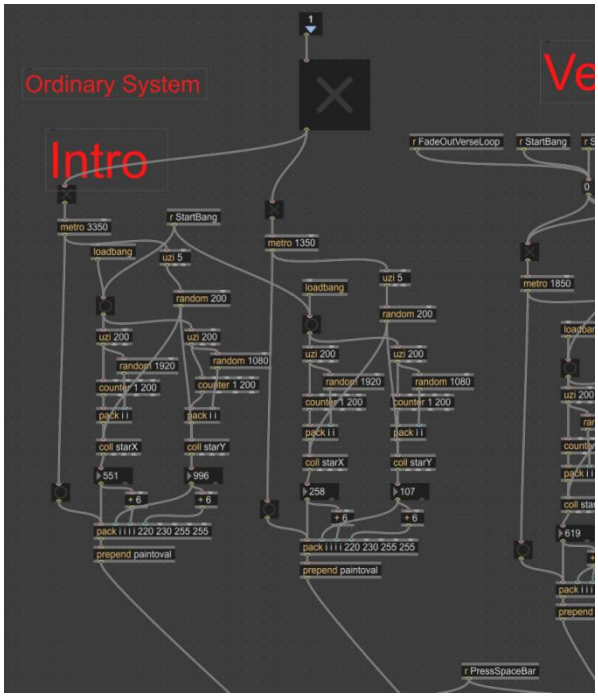


When the Boom Event appear, luminous lines emerge—binding the three objects together in that fleeting instant of unity.

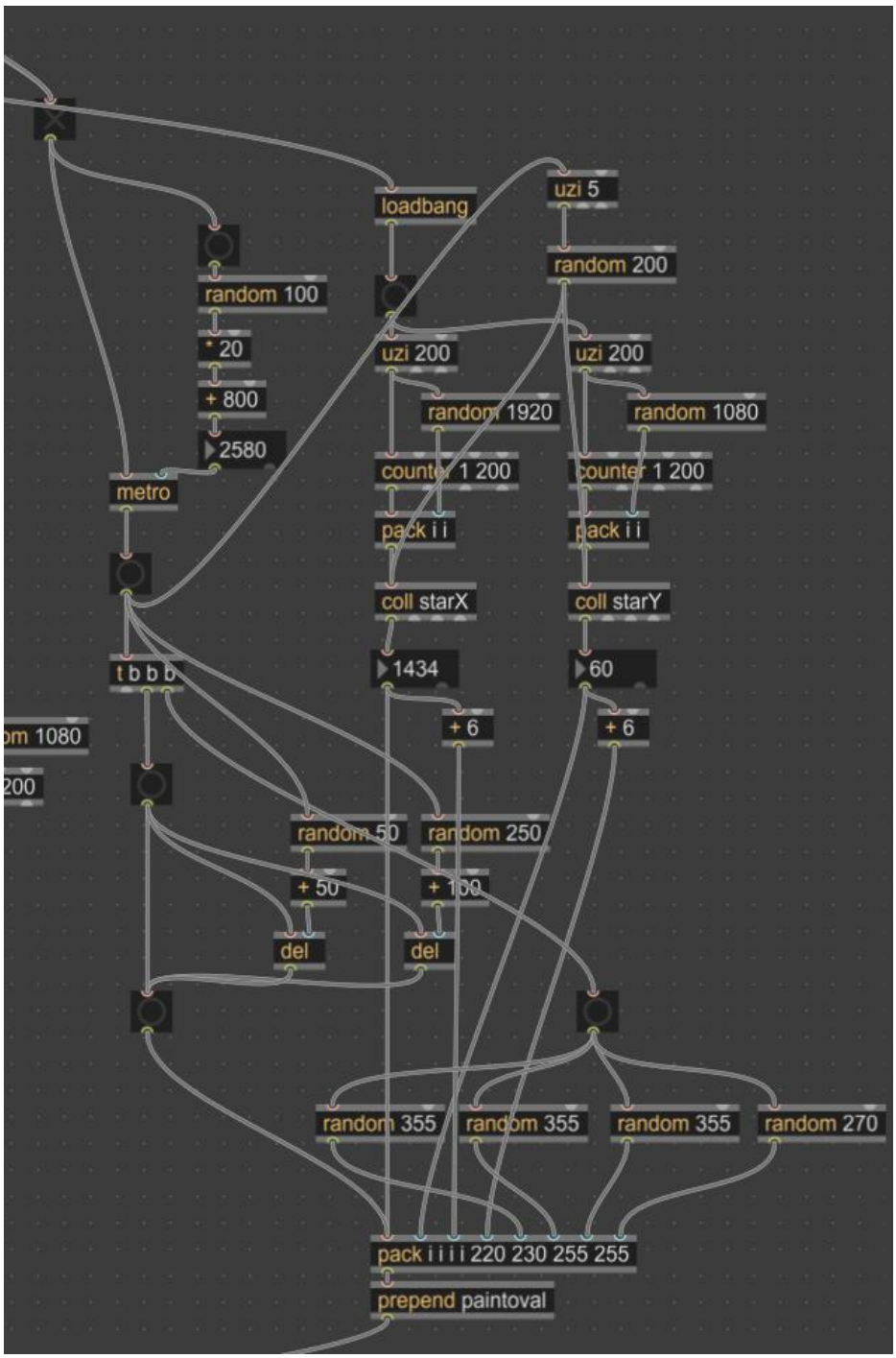
Star System



Minimum Object logic



Coloured Plate



For visual effect, I added a **Star System** that provides corresponding visual states for different progression phases.

Each algorithmic module generates **200 new (x, y) coordinate points** on every Loadbang, store them in a coll for later recall when needed.

I iterated several versions of this system, each with different point densities and configurations, to match the visual character of different process states.

Acknowledgment

Special thanks to the Sound Design teaching team for their guidance and inspiration throughout this project.